



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 3B: Flip Flops and Sequential Logic

Flip Flops (Registers) – Part I

- simple edge-triggered D type Flip-Flop

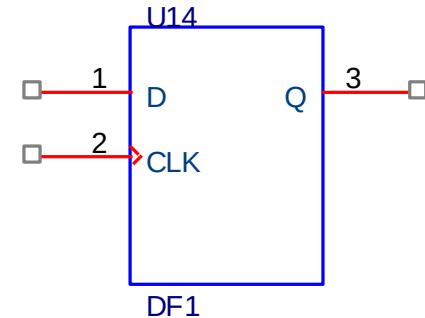
```
process (clk) -- Nothing else in sensitivity list
begin
    if(rising_edge(clk)) then
        Q <= D;
    end if;
end process;
```

- Why is D not in the sensitivity list?
 - Is it wrong to include D in the sensitivity list?
 - It's not wrong, but it's not correct!

- Alternatively

```
process (clk)
begin
    if(clk='1' and clk'event) then
        Q <= D;
    end if;
end process;
```

- Synthesizer recognizes this **template** and will happily build a DFF



Flip Flops (Registers) – Part II

- simple edge-triggered D type Flip-Flop

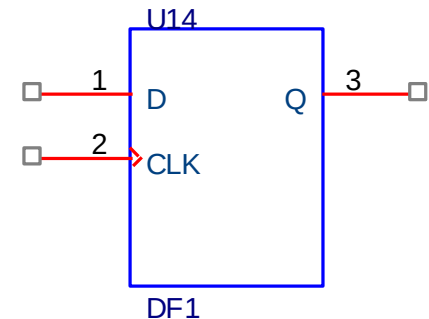
```
process (clk) -- Nothing else in sensitivity list
begin
    if (rising_edge(clk)) then
        Q <= D;
    end if;
end process;
```

- Why is D not in the sensitivity list?
 - Is it wrong to include D in the sensitivity list?
 - It's not wrong, but it's not correct!

- Alternatively

```
process (clk)
begin
    → if (clk='1' and clk'event) then
        Q <= D;
    end if;
end process;
```

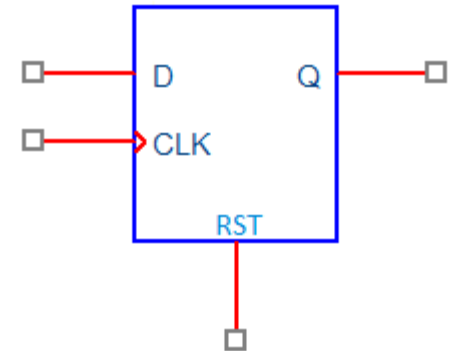
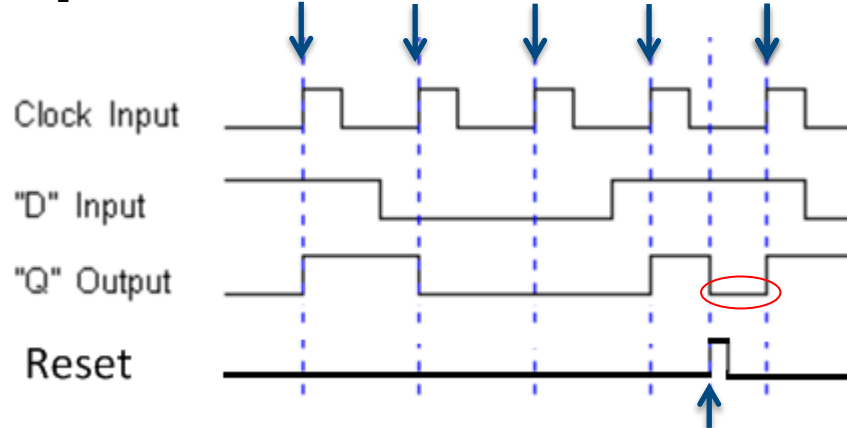
- Synthesizer recognizes this **template** and will happily build a DFF



D Type – Flipflop Variants – Part I

- DFF with **asynchronous** reset (active high reset)

```
process (clk, reset)
begin
    if (reset = '1') then
        Q <= '0';
    elsif (rising_edge(clk)) then
        Q <= D;
    end if;
end process;
```



- How do you implement a DFF with a **synchronous** clear?

D Type – Flipflop Variants – Part II

- DFF with active high reset and enable

```
process(clk, reset)
begin
    if (reset = '1') then
        Q <= '0';
    elsif (rising edge(clk)) then
        if (e = '1') then
            Q <= D;
        end if;
    end if;
end process;
```

- Why is e not in the sensitivity list?
- Notice no **else** statement for the enable
- Timing diagram?
 - Potential Quiz or Exam Question